



Encryption/Decryption Tool

Group - 08

Course Code : CSE721

Course Title : Introduction to Cryptography

Section : 01

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1. Introduction :

In this project, we have developed a cryptography tool capable of encrypting and decrypting messages using classical encryption algorithms: Caesar Cipher, Affine Cipher, Playfair Cipher and Hill Cipher. The tool is designed with a user-friendly console interface, enabling users to select the desired cipher and operation, input plaintext or ciphertext, and apply the relevant encryption or decryption process. As part of the second section of the project, we also implemented a crypto cracker tool to perform known plaintext attacks on the Hill Cipher.

The part 01 consists of four different encryption algorithms such as caesar cipher , affine cipher, play fair cipher and Hill cipher. Each of the cipher algorithms have both encryption and decryption part and also the logic behind the algorithms.

The part 02 consists of the known plaintext attack of the Hill Cipher where the plaintext and the corresponding ciphertext is known and we need to determine the key using some cryptographic formula.

2. Tools:

We have used C++ programming language and codeblocks 25.03 for writing and demonstrating the crypto tool code.

2. GitHub Link :

Here is the github link. The repository has detailed instructions to run the code .

Link : <https://github.com/tareq056/CSE721-Crypto-Tool>

Architecture

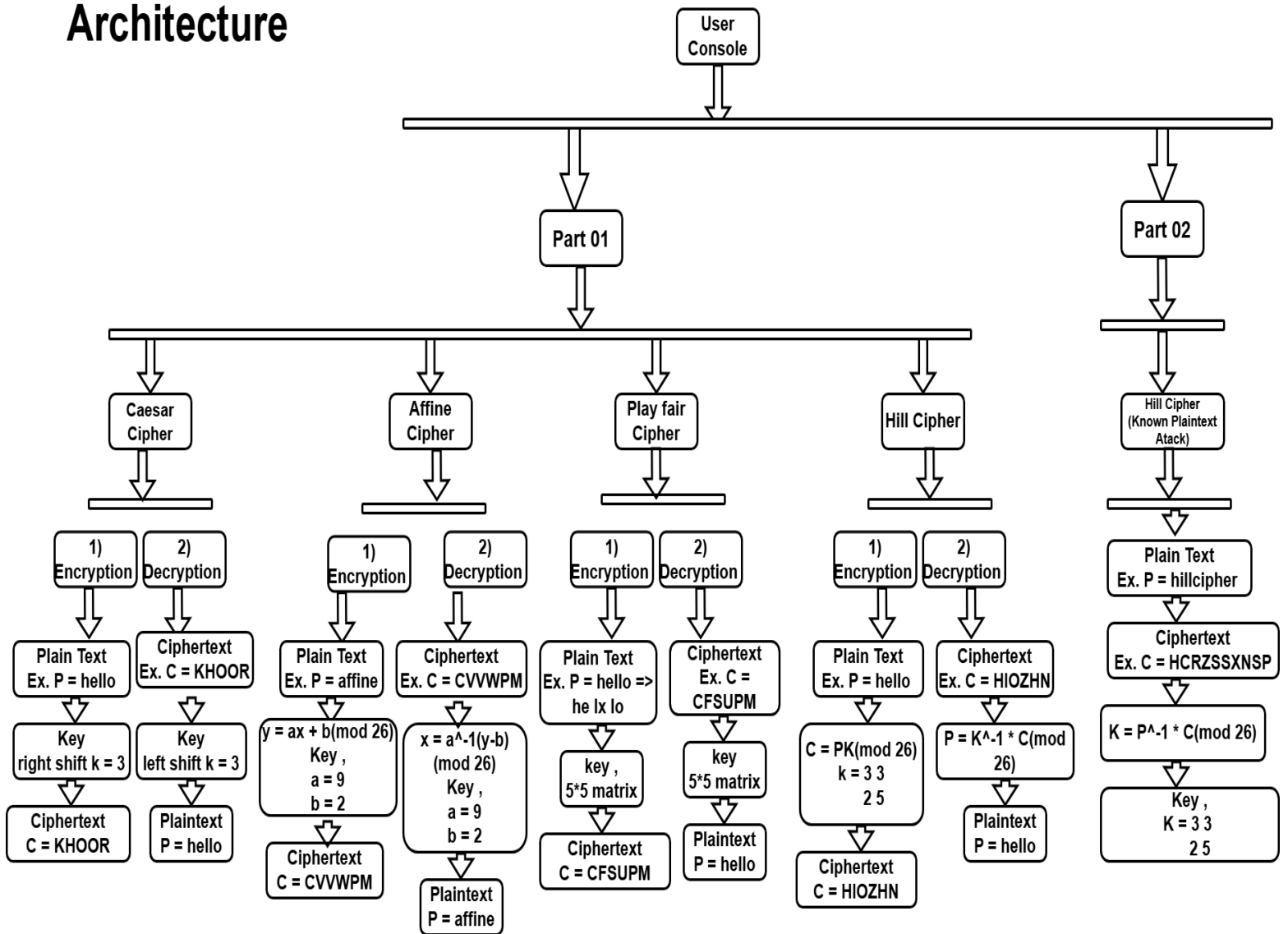


Fig. Figure of the Crypto Tool

3. Cipher Operations:

Part 01 :

Caesar Cipher:

It is basically a shift cipher. Given a plaintext and a key it does the following operations.

1) **Encryption:**

For encryption of a plaintext to make it ciphertext, each letter of the plaintext is shifted to the right of key times.

$$E(x) = (x+k) \bmod 26$$

2) **Decryption:**

For decryption of the ciphertext to make it a plaintext, each letter of the ciphertext is shifted to the left of key times.

$$D = (x-k) \bmod 26$$

Example:

Encryption	Plaintext	hello
	Key	3
	Ciphertext	KHOOR

Decryption	Ciphertext	KHOOR
	Key	3
	Plaintext	hello

```
--- Caesar Cipher ---  
1) Encrypt  
2) Decrypt  
3) Back  
Choose: 1  
Enter text: Hello  
Enter shift value: 3  
Encrypted Text: KHOOR
```

```
--- Caesar Cipher ---  
1) Encrypt  
2) Decrypt  
3) Back  
Choose: 2  
Enter text: KHOOR  
Enter shift value: 3  
Decrypted Text: hello
```

Affine Cipher:

The Affine cipher is a monoalphabetic substitution cipher where each plaintext letter maps to exactly one ciphertext letter that is built using a simple linear function over modular arithmetic. The Affine Cipher applies a mathematical function for encryption and decryption.

$$y = ax + b \pmod{26}$$

- o The above is known as the affine function. Where a and b are integers and $\gcd(a, 26) = 1$.
- o The keys here are a and b .

1) Encryption:

For encryption of plaintext to make it ciphertext, each letter undergoes the affine function.

$$E(x) = ax + b \pmod{26};$$

- o $E(x)$ is the encrypted letter
- o x is the plaintext letter
- o a and b are both the keys.

Example:

Encryption	Plaintext	affine
	Key	$a = 9, b = 2$
	Ciphertext	CVVWPM

2) Decryption:

For decryption of the ciphertext to make it plaintext, each letter of the ciphertext undergoes the following function.

- o Since $\gcd(a, 26) = 1$, so a has a multiplicative inverse mod 26.
- o For decryption, $a^{-1}a = 1 \pmod{26}$

$$D(y) = a^{-1}(y-b)$$

- o $D(y)$ is the decrypted letter
- o y is the ciphertext
- o a and b are the keys

Example:

Decryption	Ciphertext	CVVWPM
	Key	$a = 9, b = 2$
	Plaintext	affine

```

--- Affine Cipher ---
1) Encrypt
2) Decrypt
3) Back
Choose: 1
Enter text: affine
Enter key a: 9
Enter key b: 2
Encrypted Text: CVVWPM

--- Affine Cipher ---
1) Encrypt
2) Decrypt
3) Back
Choose: 2
Enter text: CVVWPM
Enter key a: 9
Enter key b: 2
Decrypted Text: affine

```

Playfair Cipher:

The Playfair algorithm is based on the use of 5*5 matrixes of letters constructed using a keyword.

- Repeating plaintext letters that are in the same pair are separated with a filler letter, such as x.
- If the two letters are in the same row, they are replaced by the letters immediately to the right, wrapping around if necessary.
- If the two letters are in the same column, they are replaced by the letter immediately below, again wrapping around if necessary.
- If the letters form a rectangle in the matrix, they are replaced by the letters in the same row but in the opposite corners of the rectangle.
- The decryption uses the same matrix but in the reverse direction order.

Example:

Encryption	Plaintext	Hello => he lx lo
	Key	monarchy
	Ciphertext	CFSUPM

Decryption	Ciphertext	CFSUPM
	Key	monarchy
	Plaintext	helxlo => hello

Key:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

--- Playfair Cipher ---

1) Encrypt

2) Decrypt

3) Back

Choose: 1

Enter key: monarchy

Enter plaintext: hello

Ciphertext (no spaces): CFSUPM

Ciphertext (digraphs): CF SU PM

--- Playfair Cipher ---

1) Encrypt

2) Decrypt

3) Back

Choose: 2

Enter key: monarchy

Enter ciphertext: CFSUPM

Plaintext: helxlo

Hill Cipher:

The Hill Cipher is a polygraphic substitution cipher that encrypts messages by applying linear algebra to groups of letters.

1. Key Matrix

- The Hill cipher uses a square matrix as the key for encryption. This matrix, usually denoted as M , has dimensions $n \times n$, where n is the number of characters in the plaintext that will be grouped together in each encryption step.
- For example, if $n = 2$, the matrix will be a 2×2 matrix.
- The key matrix is constructed using a key phrase. The letters of the key phrase are mapped to their corresponding positions in the alphabet ($A = 0, B = 1, C = 2, Z = 25$).

2. Plaintext

- The plaintext is split into groups of n letters, where n is the size of the key matrix.
- If the number of letters in the plaintext is not a multiple of n , padding is added to complete the group.

3. Matrix Representation of Plaintext

- Each group of n letters is represented as a column vector of length n . For example, if using a 2×2 Hill cipher and the group is "HELLO", then you would split it into pairs (e.g., "HE", "LL", "O").
- Each letter is converted into a numerical value using the following mapping: $A = 0, B = 1, C = 2, Z = 25$.

4. Matrix Multiplication

- The encryption process involves multiplying the key matrix with the plaintext vector.
- Let the key matrix K be a 2×2 matrix, and the plaintext vector P be a 2×1 vector.
 - $C = P * K \pmod{26}$
- This operation is performed modulo 26. The result is another column vector that represents the encrypted ciphertext.

5. Inverse Matrix for Decryption

- To decrypt the ciphertext, we need to calculate the inverse of the key matrix, K^{-1} .
 - $P = K^{-1} * C \pmod{26}$

- The inverse matrix K^{-1} can be found using the modular inverse of the determinant of K modulo 26. If the determinant of the key matrix K is not coprime with 26, the Hill cipher will not be usable.

6. Key Properties

- The key matrix must be invertible modulo 26 for decryption to work. This is a critical requirement because if the determinant of the key matrix and 26 are not coprime, the matrix does not have an inverse in the modular system, making the cipher unbreakable.
- A square matrix ensures that the system can be inverted and that each letter group ciphertext has a unique corresponding plaintext letter group.
-

Encryption	Plaintext	hello					
	Key	<table><tr><td>3</td><td>3</td></tr><tr><td>2</td><td>5</td></tr></table>		3	3	2	5
	3	3					
2	5						
Ciphertext	HIOZHN						

Decryption	Ciphertext	HIOZHN					
	Key	<table><tr><td>3</td><td>3</td></tr><tr><td>2</td><td>5</td></tr></table>		3	3	2	5
		3	3				
2	5						
Plaintext	hello						

```
--- Hill Cipher (2x2) ---
1) Encrypt
2) Decrypt
3) Back
Choose: 1
Enter plaintext: hello
Enter 2x2 key matrix (row-wise, 4 integers): 3 3 2 5
Encrypted Text: HIOZHN

--- Hill Cipher (2x2) ---
1) Encrypt
2) Decrypt
3) Back
Choose: 2
Enter ciphertext: HIOZHN
Enter 2x2 key matrix (row-wise, 4 integers): 3 3 2 5
Decrypted Text: hellox
```

Part 02:

Known Plaintext attack of Hill Cipher:

A Known Plaintext Attack is a cryptanalysis method where the attacker has access to some plaintext and the corresponding ciphertext for the same encryption key system.

Using these matched pairs, the attacker tries to recover the secret key or build a method to decrypt other messages encrypted with the same key.

In the hill cipher , it uses a key matrix with size n.

- Encryption : $C = PK \pmod{26}$

If the attacker knows enough plaintext blocks P and matching ciphertext blocks C , they can build the key matrix

- Key : $k = C * P^{-1} \pmod{26}$

Hill Cipher is very sensitive to known plaintext attacks.

Example:

Lets consider we know the plaintext is "hillcipher" and the corresponding ciphertext is "HCRZSSXNSP" .

- We choose the first two ,
 - $(7\ 8) K \pmod{26} = (7\ 2)$
 - $(11\ 11) K \pmod{26} (17\ 25)$

Using the first two plaintext-ciphertext pairs we can find the key.

```
1) Recover Key (Known Plaintext Attack)
2) Decrypt Ciphertext using Recovered Key
3) Encrypt Plaintext using Recovered Key (Verify)
4) Exit
Choose: 1
Enter known plaintext: hillcipher
Enter corresponding known ciphertext: HCRZSSXNSP
Recovered Hill Cipher Key Matrix
3 2
8 5
```

- 1) Recover Key (Known Plaintext Attack)
- 2) Decrypt Ciphertext using Recovered Key
- 3) Encrypt Plaintext using Recovered Key (Verify)
- 4) Exit

Choose: 2

Enter ciphertext to decrypt: HCRZSSXNSP

Decrypted Text: hillcipher

=== Part 2: Hill Cipher Known-Plaintext Cracker ===

- 1) Recover Key (Known Plaintext Attack)
- 2) Decrypt Ciphertext using Recovered Key
- 3) Encrypt Plaintext using Recovered Key (Verify)
- 4) Exit

Choose: 3

Enter plaintext to encrypt: hillcipher

Encrypted Text: HCRZSSXNSP